Arhaan Wazid

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Education

Bachelor of Computer Science (Honours) | *Carleton University; GPA:* **3.9/4** Sept. 2022 – April 2027

- Relevant Courses: Abstract Data Structures and Analysis of Algorithms, C Systems Programming, C++ and Java Object Oriented Programming, Database Management Systems, Web Development, Operating Systems, Programming Paradigms
- Awards and Honours: Deans' Honor List 2022-23, 2023-24, 2024-25, 2025-26 and Academic Excellence Entrance Scholarship

Experience

Embedded Software Engineer Intern | *Ciena, Ottawa, ON* Sept. 2025 – Present

- Built an internal **Common Vulnerabilities and Exposures (CVE) detection tool** in **Python** to automate vulnerability scanning across kernel versions deployed on networking hardware (e.g., **RLS 6500**), filtering and triaging CVEs per card variant
- Performed **Linux kernel upgrades** for networking hardware, resolving complex merge conflicts to integrate upstream security patches and CVE fixes without breaking compatibility with existing product code or kernel configurations
- Developed a Python tool to automate **Yocto** recipe dependency resolution and repository management, while configuring **BitBake** recipes (**.bbappend**, **.inc**) for product and simulator builds in collaboration with **DevOps**
- Served as primary point of contact for firmware updates, developing and refining **Bash** and **Python** scripts to streamline hardware flashing across embedded devices
- Debugged production **C/C++** using **Coverity** static analysis, fixing memory leaks and resource hazards in embedded Linux

Systems Software Developer Intern | *Ford Motor Company, Ottawa, ON* May 2025 – Aug. 2025

- Designed and implemented embedded **C++** components for **Android-based** infotainment systems based on a **Linux** kernel platform, integrating with low-level drivers and using **Google Test (gtest)** for unit testing to ensure reliability and performance
- Developed automated system-level infotainment tests in **Python**, increasing reliability and reducing manual testing
- Validated Enhanced Connectivity Gateway and Telematics Control Unit to ensure stable data exchange within the infotainment platform, using **TestRail** for test management, **PuTTY** for hardware verification, and **Git/GitHub** for version control

Teaching Assistant | *Carleton University, Ottawa, ON* Sept. 2023 – Jan. 2024 | Jan. 2025 – Apr. 2025

- Assisted in **Java and Python Object-Oriented Programming (OOP)** class, supporting student learning
- Conducted tutorials and office hours, clarifying key **OOP** topics like **abstract classes** and **polymorphism**
- Supported students on the class discussion board by debugging **Java and Python** code and explaining design principles

Software Developer Intern | *Ford Motor Company, Ottawa, ON* Sept. 2024 – Dec. 2024

- Automated **40+** functional tests for in-vehicle infotainment and network systems using **Python** and the **Slash** framework, boosting efficiency and increasing testing coverage by **70%**
- Managed in-vehicle infotainment system setup and test environment configuration on a **Linux operating system (Ubuntu)**, ensuring accurate connections and PC settings for outcome verification by the Stability Monitor team
- Conducted manual and automated tests to track system failures; debugged issues in **C/C++** and raised **JIRA** tickets

Projects

t:slim Insulin Pump | C++, QT, Qt Creator, Ubuntu VM

- Built a real-time **Qt-based** GUI to simulate insulin delivery, including modular components for bolus, CGM, and safety controls
- Implemented **Observer** and **Mediator** design patterns using **Qt signals/slots** to coordinate logic and UI interactions
- Developed extensible, testable code following **object-oriented principles**, enabling reusable modules across **5+** classes

Multi-thread Ghost Simulator | C, Ubuntu VM, Linux Terminal

- Collaborated on a multithreaded ghost-hunting simulator to improve performance and responsiveness on **Linux**
- Utilized dynamic data structures and **Valgrind** to optimize memory usage and resolve potential leaks
- Wrote **30+** modular functions, ghost action logs, unit tests, and **Makefile**-based builds

Skills

Languages: Java, C/C++, Python, SQL, JavaScript, HTML/CSS, JSON, Bash, Scheme

Frameworks & Technologies: Node.js, JavaFX, Yocto, Slash, PostgreSQL, RESTful APIs, MongoDB, Qt, Google Test, JUnit

Developer Tools: Git, VS Code, Testrail, Jira, PyCharm, IntelliJ, Jenkins, Linux/Unix, PuTTY, Valgrind, BitBake, Coverity

Soft Skills: Adaptability, Leadership, Time Management, Teamwork, Communication